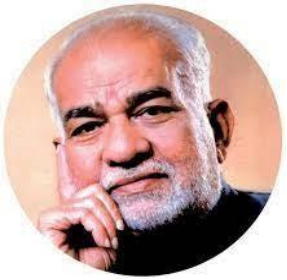
BVRIT HYDERABAD College of Engineering for Women

(UGC Autonomous)

R&D SHOWCASE 2026



URBAN ATLAS

ABSTRACT

Urban Atlas is a deep learning-based system that segments satellite images to identify land-use regions like buildings, roads, water, vegetation, and barren land. It uses a YOLO26 model for pixel-level classification and generates segmentation masks. The system provides land distribution insights through charts and evaluates performance using precision, recall, and mAP, displaying results on an interactive dashboard.

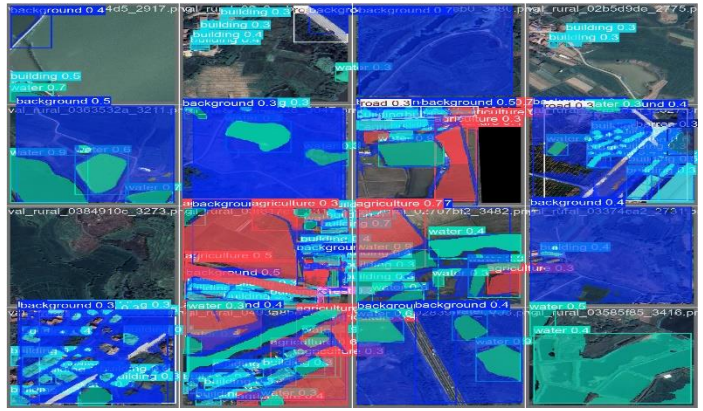
UNIQUENESS

• Pixel-level Segmentation	• Real Time Analysis
• Interactive Dashboard	• Area Distribution Insights

METHODOLOGY

The methodology uses the YOLO26 segmentation model for land-use classification. Satellite images are processed and passed through convolutional layers for feature extraction. The model performs pixel-level classification to generate segmentation masks for different land categories. Iterative training improves accuracy and performance. The system is deployed using Streamlit, enabling real-time analysis and visualisation through an interactive web interface.

Land-use Segmentation



Results & Analysis

SOCIETAL USE

The system supports SDG-11 (Sustainable Cities and Communities) by enabling efficient urban planning and environmental monitoring through satellite image analysis. It helps in understanding land-use patterns, managing resources, and making better decisions for sustainable development and smart city planning.

CONCLUSION

In conclusion, the Urban Atlas system provides an efficient solution for automatic land-use segmentation using satellite images. The YOLO26 model enables accurate and fast classification, and the web-based interface allows easy access and real-time analysis for urban planning and environmental monitoring.

REFERENCES

[1] I. Dimitrovski, V. Spasev, S. Loshkovska, and I. Kitanovski, “U-net ensemble for enhanced semantic segmentation in remote sensing imagery,” Remote Sensing, vol. 16, no. 12, p. 2077, 2024.

[2] Ultralytics, “Yolo26 model documentation,” 2024

SDG -11

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